SEMESTER 2 · B & C

ANSWERING TECHNIQUE

Short question, complete answer. State the condition you apply, then show every step.

1 HOW MARKS ARE GIVEN

1. **Formula** (marks even if substitution slips)
2. **Substitution**
3. **Answer + unit**, correct s.f.

EXAMPLE · CAPACITANCE

Discharge to 37% in 0.75 s through 1.8 MΩ:

$0.37 = e^{-t/RC} \rightarrow C = \dfrac{0.75}{1.8 \times 10^6 \times 0.994}$

$= 4.19 \times 10^{-7}$ F

EXAMPLE · DRIFT VELOCITY

$I = nAev \rightarrow v = \dfrac{I}{nAe} = 1.50 \times 10^{-4}$ m s^{-1}

Use **ε_0 from the data list**, not k = 9 × 10⁹: a slightly different answer can be marked wrong.

2 SIG. FIGS, Π & ECF

- Least s.f. in the data (STPM accepts up to 4 s.f.).
- a.c. equations in **numbers**, not π.
- **ECF:** a wrong (a) used correctly in (b) still scores.

EXAMPLE · A.C. EQUATION

× $V = 325\sin(100\pi t)$

✓ $V = 325\sin(314t)$ V

3 SIGNS: SCALAR VS VECTOR

F, E (vectors): magnitudes only; direction from a **positive test charge**.

V, U (scalars): substitute the **sign**, add algebraically.

EXAMPLE · TWO OPPOSITE CHARGES

−5 μC and +5 μC, midpoint: the two E add (both towards −5 μC), but $V = 0$.

4 DEFINE & EXPLAIN

Define: full sentence. **Explain:** how it happens, in order.

CAPACITANCE

× $C = Q/V$

✓ Ratio of the **charge on either plate** to the **p.d. between the plates**, $C = Q/V$.

DRIFT VELOCITY

✓ The **mean** velocity of the free electrons when a p.d. is applied. Missing "mean" = 0.

5 EXPLAIN IN SEQUENCE

Ask: **"After this, what happens next?"** Point form is fine if the order is right.

DIELECTRIC INCREASES C

1. Nuclei → negative plate, electrons → positive plate.
2. **Dipoles** form; opposing field.
3. **V decreases**, Q constant.
4. $C = Q/V$ **increases**.

MUTUAL INDUCTION

1. Current in primary → **flux**.
2. Flux **links** the secondary.
3. Current changes → **flux linkage changes**.
4. **E.m.f. induced** in the secondary.

6 STATE THE CONDITION

Brief question ("derive e/m") = complete answer with the **condition**.

Situation	Condition
Speed selector	$qE = qvB$
Accelerated	$eV = \frac{1}{2}mv^2$
Circle in B	$qvB = mv^2/r$
Hall voltage	$qE_H = qvB$
Held still	$qE = mg$

EXAMPLE · HELD STATIONARY

Sphere with the charge of 20 electrons: $qE = mg, q = 20(1.6 \times 10^{-19})$ C

HALL POLARITY

Metal → **electrons**; p-type → **holes**. Fleming's left-hand rule; label faces on a diagram.

7 CHOOSE THE FORMULA

V constant	$U = \frac{1}{2}CV^2$
Q constant	$U = \dfrac{Q^2}{2C}$
Heating	$\langle P \rangle = I_{rms}^2 R = \frac{1}{2}I_0^2 R$
Fuse	just above I_{rms}

POTENTIOMETER

Driver cell only: $V \propto l$.

With a resistance: find I in the driver circuit **first**.

Charged C connected to uncharged C = **parallel**; Q conserved.

8 SIMPLIFY CIRCUITS

1. Read the words given.
2. **Redraw**, following the current.
3. Combine series/parallel.

Cannot simplify → **Kirchhoff** (any current directions; write both laws).

Max power: $R = r$. MCQ: look for a **short circuit** first.

9 EXPLAIN WITH CONDUCTIVITY

$$\sigma = \frac{ne^2t}{m}$$

Metal heated: larger vibrations → mean free time t ↓ → σ ↓ → ρ ↑.

Semiconductor heated: n ↑ → σ ↑ → ρ ↓.

ρ does not depend on length: halve l → R halves, ρ same.

10 ANSWER THE QUESTION ASKED

"What external force…?" → give the **value**: $F = \dfrac{B^2l^2v}{R}$, not just "push".

Section C is not split into steps: plan it yourself (resolve horizontally and vertically, then solve).

EXAM TIPS

- ✓ MCQ: 17/20 or more is A territory.
- ✓ Use A-level reference books for definitions.
- ✓ Topics repeat (Hall effect again).
- ✓ Do not trust AI answers blindly: check the physics.